Coding Set 6

Problem 1: Division by Zero Exception

Write a C++ program that takes two integers from the user and performs division.

Requirements:

- Throw an exception if the denominator is zero.
- Catch the exception and display an appropriate error message.
- Otherwise, display the result.

Sample Input:

10

0

Output:

Error: Division by zero is not allowed.

Problem 2: Negative Number Exception

Write a program to calculate the square root of a number.

Requirements:

- If the user enters a negative number, throw a user-defined exception called NegativeNumberException.
- Display a meaningful error message.

Sample Input:

-25

Output:

Error: Square root of a negative number cannot be calculated.

Problem 3: Bank Account Withdrawal

Create a class BankAccount with balance as a data member.

Requirements:

- Allow the user to withdraw money.
- Throw an exception if the withdrawal amount exceeds the available balance.
- Catch the exception and display an error message.

Sample Input:

Balance: 5000

Withdraw: 7000

Output:

Error: Insufficient Balance.

Problem 4: Student Marks Validation

Write a program to accept marks of a student.

Requirements:

- Marks must be between 0 and 100.
- Throw an exception if invalid marks are entered.
- Handle the exception appropriately.

Sample Input:

120

Output:

Invalid Marks! Marks should be between 0 and 100.

Problem 5: Array Index Out of Bounds

Create an array of 10 elements.

Requirements:

- Ask the user to enter an index.
- Throw an exception if the index is less than 0 or greater than 9.
- Display the element if the index is valid.

Sample Input:

15

Output:

Error: Array Index Out of Bounds.

Problem 6: Voting Eligibility Checker

Write a program to determine voting eligibility.

Requirements:

- Accept age from the user.
- Throw a user-defined exception if age is less than 18.

- Display whether the person is eligible to vote.

Sample Input:

16

Output:

Exception: Not eligible for voting.

Problem 7: Multiple Catch Blocks

Write a menu-driven calculator.

Requirements:

- Perform Addition, Subtraction, Multiplication, and Division.
- Throw:
    - Integer exception for division by zero.
    - Character exception for invalid operator.
- Handle exceptions using multiple catch blocks.

Sample Input:

10 / 0

Output:

Division by Zero Error.

File Handling Problems (3)

Problem 8: Student Record Writer

Write a program to store student information in a file.

Requirements:

- Accept Roll Number, Name, and Marks.
- Save the details into a file named students.txt.
- Display a success message after writing.

Sample Output File:

101 Ali 87

102 Sara 92

Problem 9: Count Characters, Words, and Lines

Write a program that reads a text file and counts:

- Total number of characters
- Total number of words
- Total number of lines

Requirements:

- Read data from a file named article.txt.
- Display all three counts.

Sample Output:

Characters: 250

Words: 45

Lines: 8

Problem 10: Copy Contents from One File to Another

Write a program to copy data from one file to another.

Requirements:

- Read contents from source.txt.
- Write contents into destination.txt.
- Display a success message after copying.

Sample Output:

File copied successfully.